ANALYTIX AI COPILOT GUIDE

Natural Language Querying, Grounding, & Scientific Prompting Guide

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1. AI Scientist Copilot Operations Guide

The AI Scientist Copilot is an advanced natural language interface designed to query chemical databases, structural libraries, and assay records using grounding rules.

2. Grounding Rules & Hallucination Prevention

To prevent model hallucinations, the copilot validates all generated SQL queries against a strict schema mapping before execution. Queries that fail validation are automatically blocked.

3. Scientific Prompt Examples Library

Natural Language Prompt	Target Database Fields	Expected Query Outputs
'What compounds target EGFR?'	compounds, assay_targets, ic50	Table of active compounds and IC50 potency metrics against EGFR.
'Show compounds with MW < 500 and LogP < 5'	molecular_weight, clogp, smiles	List of small molecule candidates matching Lipinski's Rule of 5.
'Summarize assay results'	assay_runs, results, inhibition	Statistical summaries, average inhibition, and active hit counts across a
'Find compounds active against KRAS'	compounds, target_proteins, active_status	Potent candidates with low nanomolar activity against KRAS targets.
'Show workflow execution history'	workflow_definitions, workflow_runs	Log history of automated pipeline executions and runtime statistics.

4. Formatting and Exporting Copilot Data

Search results generated by the copilot can be exported directly. Use the 'Export to CSV' button on the results panel to save target compound tables and descriptors locally.